

Coding & Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/rithikasundari/Emotion-Detection-Learning-Support-Engine
Programming Language(s)	Python
Framework(s) Used	Streamlit, TensorFlow, Transformers, NLTK
Key Features Implemented	The system detects student emotions using BiLSTM and BERT models and provides personalized learning recommendations through Gemini AI. It includes a Streamlit dashboard for emotion analysis and progress tracking.
Pending / Incomplete Features	User authentication, multi-user support, advanced cloud deployment, and additional emotion categories are planned for future development.
Setup / Run Instructions	Install the required Python libraries, configure the Gemini API key, and run the Streamlit application to access the system.

Code Quality Checklist

S.N o	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The project combines NLP, Deep Learning, and Generative AI to identify student emotions and provide personalized learning support for improved academic performance